# Topic 5: Formulae, Equations and Amounts of Substance

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include determining a simple empirical formula such as MgO or CuO, determining the number of moles of water of crystallisation in a salt such as Epsom salts, performing a wide range of titrations involving different indicators, preparing salts.

Mathematical skills that could be developed in this topic include converting between units such as $cm^3$ and $dm^3$, using standard form with the Avogadro constant, rearranging formulae for calculating moles in solids and in solutions, calculating atom economy, dealing with percentage errors.

Within this topic, students first encounter core practicals and can consider ideas of measurement uncertainty, evaluating their results in terms of systematic and random errors. They can also consider how the concept of atom economy is useful to help chemists make decisions so that reactions can be made more efficient in terms of resources.

| **Students should:** |
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| 1. know that the mole (mol) is the unit for amount of a substance |
| 2. be able to use the Avogadro constant, *L*, ($6.02 \times 10^{23}$ $mol^{-1}$) in calculations |
| 3. know that the molar mass of a substance is the mass per mole of the substance in g $mol^{-1}$ |
| 4. know what is meant by the terms 'empirical formula' and 'molecular formula' |
| 5. be able to use experimental data to calculate<br>i. empirical formulae<br>ii. molecular formulae including the use of $pV = nRT$ for gases and volatile liquids<br>*Calculations of empirical formula may involve composition by mass or percentage composition by mass data*. |
| 6. be able to write balanced full and ionic equations, including state symbols, for chemical reactions |
| 7. be able to calculate amounts of substances (in mol) in reactions involving mass, volume of gas, volume of solution and concentration<br>*These calculations may involve reactants and/or products*. |
| 8. be able to calculate reacting masses from chemical equations, and vice versa, using the concepts of amount of substance and molar mass |
| 9. be able to calculate reacting volumes of gases from chemical equations, and vice versa, using the concepts of amount of substance |
| 10. be able to calculate reacting volumes of gases from chemical equations, and vice versa, using the concepts of molar volume of gases |
| ***CORE PRACTICAL 1: Measure the molar volume of a gas*** |
| 11. be able to calculate solution concentrations, in mol $dm^{-3}$ and g $dm^{-3}$, including simple acid-base titrations using a range of acids, alkalis and indicators<br>*The use of both phenolphthalein and methyl orange as indicators will be expected*. |

| Students should: |
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| ***CORE PRACTICAL 2: Prepare a standard solution from a solid acid and use it to find the concentration of a solution of sodium hydroxide*** |
| ***CORE PRACTICAL 3: Find the concentration of a solution of hydrochloric acid*** |
| 12. be able to:<br>i calculate measurement uncertainties and measurement errors in experimental results<br>ii comment on sources of error in experimental procedures |
| 13. understand how to minimise the percentage error and percentage uncertainty in experiments involving measurements |
| 14. be able to calculate percentage yields and percentage atom economies using chemical equations and experimental results<br>$\text{Atom economy of a reaction} = \dfrac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}} \times 100\%$ |
| 15. be able to relate ionic and full equations, with state symbols, to observations from simple test tube reactions, to include:<br>i displacement reactions<br>ii reactions of acids<br>iii precipitation reactions |
| 16. understand risks and hazards in practical procedures and suggest appropriate precautions where necessary |